PORTFOLIO CASE STUDY · HEALTH DATA AND ANALYTICS

Hospital Capacity in 2026: Where Beds and Flow Break Down, and Where AI Helps

A written case study, a slide deck, an interactive dashboard and an R analysis, compiled in one place. October 2026.

AT A GLANCE

What the work found

- Staffed beds fell about 16% while hospitalizations held roughly flat, lifting average adult occupancy from about 64% to about 75%. The study behind this projects 85%, its shortage threshold, by 2032 if beds and admission rates stay flat.
- Capacity breaks at the ED door (admitted patients boarding), at discharge (Florida: 781,000 unnecessary days in a year) and in the operating room (idle block time).
- The best-evidenced AI result is discharge prediction used in daily rounds: median stay down more than 12 hours on 2 of 4 units at one hospital, with no concurrent control group.
- Census forecasting, inflow tools, OR block scheduling and staffing tools rest mostly on self-reported or vendor-supplied results. A reported 35-bed saving from one vendor-sponsored source is larger than its own inputs imply (22 to 30 beds).
- On simulated data, staggered difference-in-differences recovers a known effect with about a third of the error of a naive before-and-after comparison.

DELIVERABLE 1

Written case study

The full analysis, with sources linked inline and listed at the end.

Staffed beds fell about 16% while hospitalizations held roughly flat, lifting U.S. occupancy from about 64% before the pandemic to about 75%. AI tools for census forecasting, discharge prediction, command centers and OR scheduling report real gains, but most of the evidence is self-reported or vendor-supplied.

The capacity squeeze

Staffed hospital beds fell about 16% while hospitalizations stayed roughly flat, which pushed national occupancy up. A UCLA study in JAMA Network Open, as reported by HCI Innovation Group, used CDC data covering nearly every U.S. hospital from August 2020 to April 2024 and defined occupancy as census divided by staffed beds. It treats 85% as the shortage threshold for general beds and projects the national average could reach that level by 2032. The projection assumes hospitalization rates and bed supply stay unchanged and reports no confidence intervals in the coverage I read. The study did not test why beds fell; staffing shortages and closures are the author's suggested explanations from other literature.

MEASURE	FIGURE	SOURCE AND CAVEAT
Average adult occupancy, decade before the pandemic	About 64%	UCLA study via HCI
Average adult occupancy after the pandemic	About 75%	UCLA study via HCI
Change in staffed beds	About 16% fewer	UCLA study via HCI
Projected occupancy in 2032	85%	Projection; depends on flat beds and an aging population
Admitted ED patients waiting 4+ hours for a bed	More than 25% overall; 35% in peak winter months	Health Affairs analysis of 46 million visits, via Becker's; original not opened
Admitted patients waiting 24+ hours in 2024	Nearly 5% in peak months vs about 2.6% off-peak	Same analysis

Staffing is the other half of capacity, because a bed without a nurse is not a staffed bed. The 2026 NSI report puts RN turnover at 17.6%, up from 16.4% in 2024, as quoted by registerednursing.org.

Where flow breaks

Capacity problems show up at three points, and they need different fixes.

- **The front door.** Admitted patients board in the ED until an inpatient bed opens. CMS finalized an Emergency Care Access and Timeliness measure in the CY2026 OPPS final rule that requires hospitals to track and publicly report how long admitted patients stay in the ED; ACEP's statement did not give a start date. Boarding will become a reported number, which makes it a natural outcome metric for any flow project.
- **The discharge.** Florida hospitals reported about 243,000 patients (nearly 1 in 10) who stayed after being cleared for discharge between July 2024 and June 2025, totaling about 781,000 unnecessary days and about $2.5 billion in spending, per a Florida Hospital Association survey reported by The Capitolist. Health plan approvals for follow-up care accounted for 41% of delayed patients, a shortage of skilled nursing or rehabilitation beds affected about 41%, and prior authorization was linked to 52% of delays among Medicare Advantage patients. 158,000 ED patients (7.4%) waited more than 12 hours for a hospital bed.
- **The operating room.** Reserved surgical block time that is neither used nor released sits idle, while other cases wait. Providence's results below target this directly.

The Florida numbers matter for AI claims: most listed causes sit with payers and post-acute facilities, outside the hospital's control. A discharge-prediction model can tell a team who is likely to leave tomorrow; it cannot approve a prior authorization. The survey is from a hospital association, covers one state, and I did not find an independent check.

Trend-to-solution map

The best evidence is for discharge prediction paired with a changed workflow. Command centers have a promising but uneven record, and census forecasting, OR scheduling and staffing tools rest mostly on self-reported or vendor-supplied results.

BOTTLENECK	AI APPROACH	BEST EVIDENCE FOUND	EVIDENCE QUALITY
Unexpected census swings	Admission and census forecasting	Children's Mercy reports up to 98% accuracy on two-week patient counts	Low; self-reported, baseline unstated
Late or unplanned discharges	Machine-learning discharge prediction feeding daily rounds	BMJ Innovations study: median stay down more than 12 hours on 2 of 4 units	Moderate; peer-reviewed, no concurrent control
ED crowding and bed assignment	Digital command centers with real-time analytics	Scoping review of 15 studies: boarding down by up to about 90%, stay down about 20% in some ML pathways	Low to moderate; heterogeneous designs
Inpatient throughput	Predictive inflow tool (LeanTaaS iQueue)	UCHealth: stay down 0.4 days, equal to 35 beds	Low; vendor webinar, method not shown
Idle surgical block time	AI block scheduling and release	Providence: 30% less unused and unreleased block time	Low; baseline not stated
Staffing as a bed limit	Predictive staffing and scheduling	Deloitte survey via DXC: 56% of executives expect operational value	Low; expectation, not outcome

Evidence in detail

Discharge prediction

This is the most rigorous result I found. A BMJ Innovations study at a community hospital in Columbia, Maryland used unit-specific random forest models, refreshed three times a day, to predict discharges on four adult inpatient units. A case manager presented each prediction in multidisciplinary rounds, and the team set a target date in about two minutes per patient. The

analysis used interrupted time series with segmented regression on 15,613 patients before and 12,470 after rollout, with units going live one at a time.

- Median stay fell by more than 12 hours on one medicine unit ($p<0.001$) and the telemetry unit ($p=0.002$).
- The second medicine unit and the surgical unit showed no significant change.
- Prospective AUC was about 0.70 to 0.78 for same-day and next-day predictions: useful, not sharp.
- Staff received about 132 hours of just-in-time support, and the authors report that physician absence, case manager turnover and inconsistent whiteboard use affected rounds.

Two cautions. There was no concurrent control unit and no randomization, so confounding cannot be ruled out, and it is one hospital. The source also disagrees with itself in places (AUC and specificity ranges differ between abstract and results, and so do the study dates), so I used the results-section figures. The practical lesson is that the model alone did not shorten stays; the rounds workflow around it did, and it worked unevenly.

Command centers

A scoping review of 15 studies from 2021 to 2025 (47% quantitative, 20% qualitative, 33% mixed methods) found quantitative reports of boarding reductions up to roughly 90%, about 20% shorter stays in some ML-supported pathways, and several sites keeping occupancy under about 85%. Mixed-methods evaluations found length-of-stay effects were inconsistent and benefits varied across subgroups. The authors note few high-quality trials, no standard definition of a command center, and data quality and training burdens. The upper-end numbers should be read as best cases from different designs, not an expected effect.

Forecasting versus outcomes

Forecast accuracy is not an operational outcome. A model that predicts tomorrow's census as well as "same as last week" adds little, so the useful test is error against a simple baseline and then a change in boarding hours, length of stay or diversions. That is the missing piece in the Children's Mercy claim. A simulation study at a Shenzhen hospital estimated about 30% shorter outpatient waits from AI scheduling, but that is a model of a hospital, not a deployment (Frontiers in AI).

Operating room block time

Providence, a 51-hospital system, reports a 30% reduction in unused and unreleased OR block time after rolling out LeanTaaS iQueue for Operating Rooms. Its early-release window grew from under seven days to up to 28 days, and 34% of released slots were backfilled, with

the share rising month over month. The Becker's interview does not give the baseline period or the calculation, and the sites went live in phases, so I treat the 30% as a claim to verify.

Capacity outside the building

Hospital at home adds inpatient-level capacity without adding beds. According to Becker's ASC, citing the AMA, the waiver was extended through 2030 under the 2026 Consolidated Appropriations Act, and 366 programs across 139 health systems in 37 states have been approved. That is a policy and operating change, not an AI result. I found no controlled evidence on AI for patient selection or monitoring in these programs, so I list it as context.

How much weight each source deserves

SOURCE TYPE	EXAMPLES IN THIS PAPER	HOW I TREAT IT
Peer-reviewed, quasi-experimental	BMJ Innovations discharge prediction study	Primary evidence; one hospital, no concurrent control
Peer-reviewed projection	UCLA occupancy study (via HCI)	Sound for direction; the 2032 figure depends on assumptions
Peer-reviewed analysis, secondary report	Health Affairs boarding analysis (via Becker's)	Credible data, but I did not open the original
Scoping review	Command center review	Shows the range of claims, not a pooled effect
Hospital self-report	Children's Mercy, Providence	Plausible, unverified; no baseline stated
Vendor-sponsored	UCHealth webinar	Hypothesis to test, not a result
Industry or association survey	Florida Hospital Association, Deloitte	Describes conditions or expectations, not AI outcomes

Measurement plan

This is how I would test whether each AI use improves capacity, using the weaknesses found above: unstated baselines, no comparison groups, and metrics that can be gamed.

USE CASE	PRIMARY METRIC	COMPARISON DESIGN	GUARDRAIL METRICS
Census forecasting	Error (MAE or WAPE) at 24-hour, 7-day and 14-day horizons against a same-week-last-year baseline	Rolling-origin backtest, then a prospective run	Bias by day of week; error during surges; diversions and overtime as decision outcomes
Discharge prediction	Risk-adjusted observed over expected length of stay; avoidable days; share of discharges before noon	Staggered unit rollout with difference-in-differences against not-yet-live units; dose-response by rounds participation	30-day readmission; return ED visits; mortality
ED boarding and command centers	Hours from admit order to ED departure per admitted patient; share waiting 4+ and 24+ hours	Interrupted time series with a respiratory-season term; compare with non-adopting peer hospitals	Left-without-being-seen rate; mortality; staff overtime
OR block scheduling	Unused plus unreleased block minutes as a share of allocated; release lead time; backfill rate	Pre and post with same-season comparison; unaffected services or sites as controls	Late starts; turnover time; surgeon cancellations; access by service
Staffing and scheduling	Shifts filled against target; staffed-bed days; agency spend	Staggered rollout by unit	Turnover; overtime; nurse-to-patient ratio

Three cautions apply throughout. Occupancy depends on the denominator, so say whether it uses staffed or licensed beds. Length of stay and boarding both follow season and case mix, so any before-and-after claim needs a control for both. And pushing earlier discharges can raise readmissions, so no discharge metric should be reported without its guardrail.

Sources

Pages opened in full:

- U.S. Hospitals Facing Potential Bed Shortages, UCLA Researchers Project, HCI Innovation Group, February 19, 2025 (secondary report of a JAMA Network Open study; original not opened)
- ED boarding in 2025: 4 notes, Becker's, December 29, 2025 (secondary report of a Health Affairs analysis; original not opened)
- ACEP statement on the CMS 2026 OPPS final rule and the new ED boarding measure, November 21, 2025
- Report: 243,000 Florida patients faced delays leaving hospitals, The Capitolist, July 2026
- Machine-learning-based hospital discharge predictions, BMJ Innovations
- The effects of digital hospital command centers on emergency department patient throughput and crowding: scoping review
- Inside Providence's 30% reduction in unused and unreleased OR block time, Becker's
- CMS extends hospital-at-home waivers for 5 years, Becker's ASC
- Children's Mercy uses AI to predict hospital surges, August 12, 2026
- UCHealth Streamlines Inpatient Flow with AI, AHA sponsored webinar, 2023

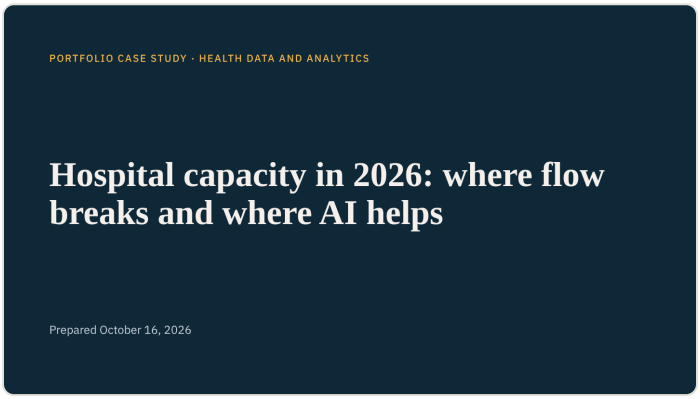
Cited from search-result excerpts only (not opened in full; verify before relying on them):

- RN turnover figures, registerednursing.org
- Deloitte survey via DXC
- Shenzhen outpatient simulation, Frontiers in AI

DELIVERABLE 2

Slide deck

The twelve slides as presented. Each caption gives the slide's title.



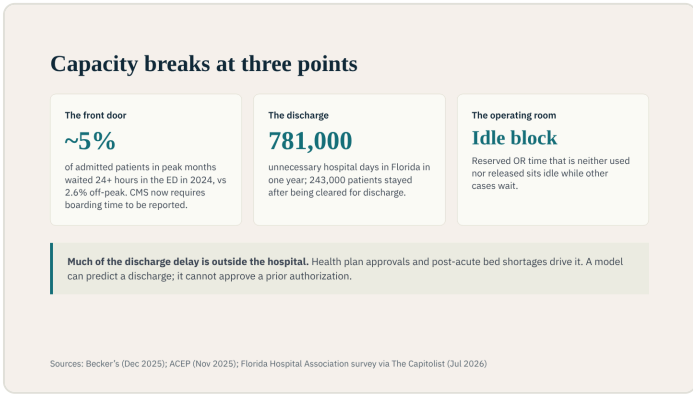
Slide 1 · Cover



Slide 2 · The short version



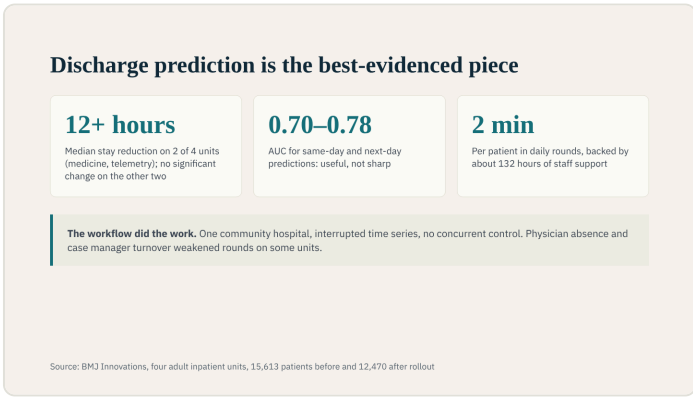
Slide 3 · Occupancy rose because staffed beds fell



Slide 4 · Capacity breaks at three points



Slide 5 · Where AI is working, and how well we know it



Slide 6 · Discharge prediction is the best-evidenced piece

Command centers: strong best cases, uneven length-of-stay effects

15
Studies in the scoping review, 2021 to 2025

~90%
Largest reported reduction in ED boarding

~20%
Shorter stay in some ML-supported pathways

Treat the upper end as a best case. Mixed-methods evaluations found length-of-stay effects inconsistent. The authors note few high-quality trials and no standard definition of a command center.

Source: scoping review of digital hospital command centers and ED throughput

Slide 7 · Command centers: strong best cases, uneven effects

Self-reported claims to test, not accept

Census forecast
98%
Children's Mercy: accuracy predicting patient numbers over two weeks. Baseline and outcome change not stated.

Inpatient flow
0.4 days
UCHealth: length-of-stay cut, equal to 35 beds. From a 2023 vendor webinar; method not shown.

OR block time
30%
Providence: less unused and unreleased OR block time. Baseline period and calculation not given.

Accuracy is not an outcome. Test a forecast against a naive baseline such as same-week-last-year, then look for change in boarding hours, length of stay and diversions.

Sources: Children's Mercy (Aug 12, 2026); AHA sponsored webinar page (2023); Becker's interview with Providence

Slide 8 · Self-reported claims to test, not accept

Not every source deserves equal weight

Source type	Examples here	How I treat it
Peer-reviewed, quasi-experimental	BMJ Innovations discharge study	Primary evidence; one hospital, no control
Peer-reviewed projection	UCLA occupancy study	Direction is sound; 2032 depends on assumptions
Secondary report of an analysis	Health Affairs boarding data	Credible; original not opened
Scoping review	Command center review	Range of claims, not a pooled effect
Hospital self-report	Children's Mercy, Providence	Plausible but unverified; no stated baseline
Vendor-sponsored	UCHealth webinar	A hypothesis to test, not a result
Association or industry survey	Florida Hospital Association, Deloitte	Conditions and expectations, not AI outcomes

Slide 9 · Not every source deserves equal weight

How I would test whether it works

Use case	Primary metric	Comparison design	Guardrails
Census forecasting	Error vs same-week-last-year baseline at 24-hour, 7-day, 14-day horizons	Rolling-origin backtest, then prospective run	Bias by weekday; error during surges
Discharge prediction	Observed over expected length of stay; avoidable days	Staggered unit rollout, difference-in-differences	30-day readmission; return ED visits
ED boarding	Hours from admit order to ED departure; share waiting 4+ and 24+ hours	Interrupted time series with season term; peer hospitals	Left-without-being-seen; mortality
OR block	Unused plus unreleased block minutes; release lead time; backfill	Same-season pre and post; unaffected services as controls	Late starts; turnover time; cancellations

Three cautions: say whether occupancy uses staffed or licensed beds; control for season and case mix; and never report a discharge metric without its readmission guardrail.

Slide 10 · How I would test whether it works

Three takeaways for an analyst

Measure outcomes, not forecast accuracy
Beat a naive baseline first, then track boarding hours, length of stay and diversions.

Pair every model with a workflow
The discharge study's gains came from daily rounds built around the prediction, and varied by unit.

Separate what the hospital controls
Many discharge delays sit with payers and post-acute beds, so attribute effects with care.

Slide 11 · Three takeaways for an analyst

Sources

Opened in full

- HCI Innovation Group: UCLA bed shortage projection (Feb 2025)
- Becker's: ED boarding in 2025 (Dec 29, 2025)
- ACEP: CMS boarding measure statement (Nov 21, 2025)
- The Capitolist: Florida discharge delays (Jul 2026)
- BMJ Innovations: ML discharge prediction study
- Scoping review: digital hospital command centers
- Becker's: Providence OR block time
- Becker's ASC: hospital-at-home waiver extension
- Children's Mercy (Aug 2026); AHA UCHealth webinar (2023)

Search excerpts only, verify before relying

- registerednursing.org: RN turnover from the NSI report
- DXC: Deloitte survey on AI operations
- Frontiers in AI: Shenzhen outpatient simulation

The UCLA study and Health Affairs analysis were read through secondary reports; the original papers were not opened.

Full links are in the written case study.

Slide 12 · Sources

DELIVERABLE 3

Interactive dashboard

A snapshot of the dashboard. In the live version, selecting a row of the evidence table shows the evidence behind it, and each chart has hover details. The data is fixed (uploaded CSVs), not live.

Hospital capacity and AI

Average occupancy rose from about 64% to about 75% as staffed beds fell 16%. The strongest AI evidence so far is for late discharges.

Average occupancy after the pandemic
75%
Adult hospitals, UCLA study via HCI

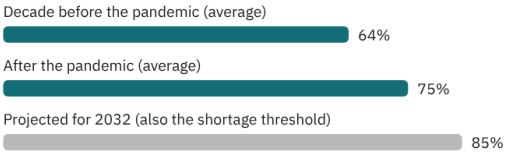
Fall in staffed beds
16%
While hospitalizations held roughly flat

Unnecessary days from discharge delays
781,000
Florida, July 2024 to June 2025

Median stay cut on the units that improved
12+ hours
Discharge prediction in daily rounds, BMJ Innovations

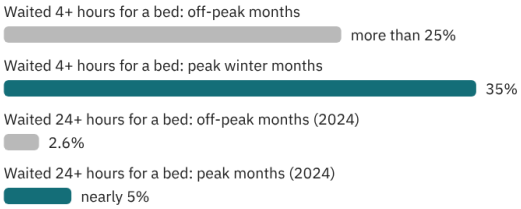
Occupancy has risen toward shortage levels

Average national adult occupancy: census divided by staffed beds. The 2032 figure is a projection that assumes beds and hospitalization rates stay unchanged.



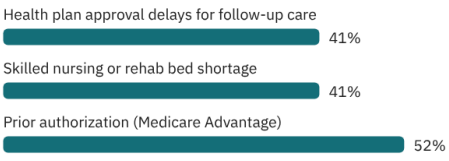
Admitted patients wait in the ED

Share of ED visits leading to admission that waited for an inpatient bed, Health Affairs analysis of 46 million visits via Becker's. Different years apply to the two thresholds.



Much of discharge delay sits outside the hospital

Causes of delayed discharge, Florida Hospital Association survey via The Capitolist. The base differs by bar, and a hospital association survey is not an independent check.



Discharge prediction helped on two of four units

Interrupted time series at one community hospital. No concurrent control unit.

Unit	Type	Patients before	Result	Significance
Medicine A	Medical	3,694	Median stay down more than 12 hours	p<0.001
Medicine B	Medical	2,945	No significant change	Not significant
Surgery C	Surgical	4,700	No significant change	p=0.190
Telemetry D	Telemetry	4,274	Median stay down more than 12 hours	p=0.002

Where AI is being applied, and how well we know it

Select a row for the evidence behind it. The evidence score is my rating from 1 (expectation, vendor or self-reported claims) to 5 (controlled, independent evidence), based on study design and who reported the result.

Bottleneck	AI approach	Evidence quality	Score
Census swings	Admission and census forecasting	Low	2
Late discharges	ML discharge prediction in daily rounds	Moderate	3
ED crowding	Digital command centers	Low to moderate	2
Inpatient throughput	Predictive inflow tool	Low	1
Idle OR block time	AI block scheduling and release	Low	2
Staffing as a bed limit	Predictive staffing and scheduling	Low	1

Census swings

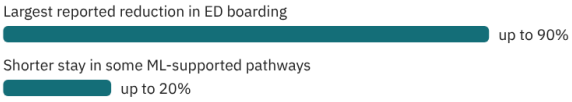
AI approach: Admission and census forecasting · Evidence quality: Low

Children's Mercy reports up to 98% accuracy on two-week patient counts, three years into its Patient Progression Hub. The accuracy metric and baseline are not stated, and no outcome change is reported.

Open source

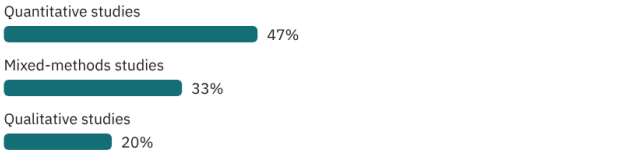
Command centers: reported best cases

Scoping review of 15 studies. These are upper-end results from different designs, not an expected effect.



Evidence base: study designs reviewed

Share of reviewed studies by design. Few high-quality trials, and no standard definition of a command center.



Sources: UCLA study via HCI Innovation Group; Health Affairs analysis via Becker's; ACEP; Florida Hospital Association survey via The Capitolist; BMJ Innovations; command center scoping review; Providence via Becker's; Children's Mercy; AHA webinar page; DXC. Full links are in the written case study.

DELIVERABLE 4

R analysis

Seven charts built with dplyr and ggplot2. Charts 1 to 5 use published figures and my own arithmetic on them. Charts 6 and 7 use simulated data to demonstrate the measurement plan and say nothing about any real hospital.

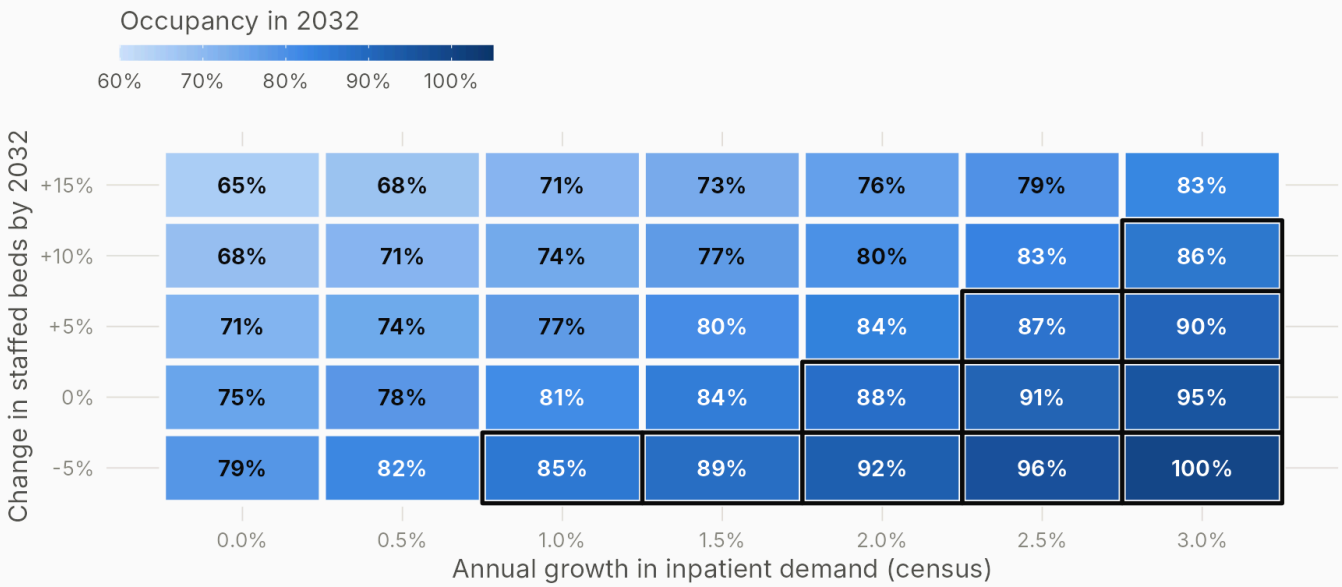
CHART 1 · PUBLISHED FIGURES

Occupancy in 2032 under different demand and bed scenarios

How far could occupancy rise if beds stay flat and demand grows?

Holding beds flat, modest demand growth reaches the 85% shortage line

Scenario model from the 75% post-pandemic average. Outlined cells reach 85%. With beds flat, the study's own 85% by 2032 implies demand growth of about 1.6% a year over 8 years.



Model: occupancy_2032 = occupancy_now x (1 + demand growth)^8 / (1 + change in staffed beds). My arithmetic, not the study's model. Baseline 75% and 85% threshold: UCLA study in JAMA Network Open via HCI Innovation Group (Feb 2025). The 8-year horizon is my assumption.

Occupancy is census divided by staffed beds, so a scenario is one line of arithmetic: 75% × $(1 + \text{demand growth})^8 \div (1 + \text{change in staffed beds})$. I checked the identity first: 64% before the pandemic divided by (1 – 16%) gives 76%, close to the 75% reported. Back-solving the study's own projection (85% by 2032, beds flat) gives demand growth of about 1.6% a year over 8 years, 2.1% over 6 years and 1.3% over 10.

> **Read with care.** The 8-year horizon is my assumption, and this is my arithmetic, not the study's model.

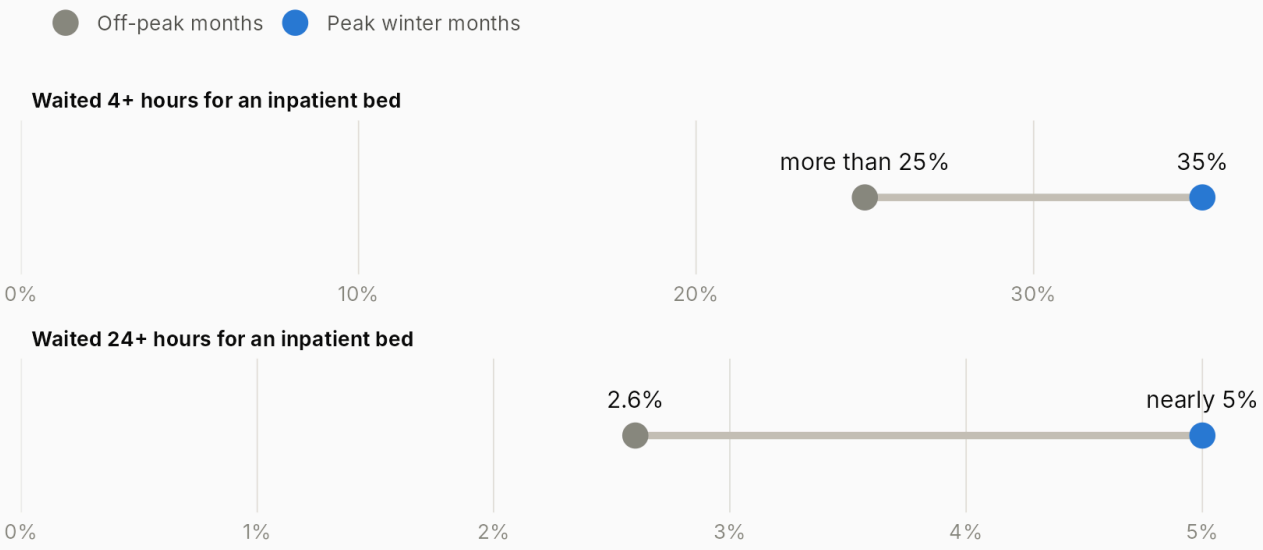
CHART 2 · PUBLISHED FIGURES

ED boarding waits, peak versus off-peak

How much worse are waits for an inpatient bed in peak winter months?

Peak winter months push more admitted patients into ED waits

Share of ED visits leading to admission, by wait for an inpatient bed. The off-peak 4-hour value is a lower bound (more than 25%) and the peak 24-hour value is rounded (nearly 5%).



Health Affairs analysis of 46 million ED visits at about 1,500 hospitals via Becker's (Dec 29, 2025); original not opened. The 4-hour shares cover the last three years and the 24-hour shares cover 2024, so compare within a panel only.

The share waiting 4 hours or more rises by 10 points, from more than 25% to 35%. The share waiting 24 hours or more rises from 2.6% to nearly 5%, about 1.9 times as many. Each threshold has its own panel because the scales differ by a factor of ten.

> **Read with care.** The off-peak 4-hour value is a lower bound and the peak 24-hour value is rounded, so both differences are approximate. The 4-hour and 24-hour figures cover different years.

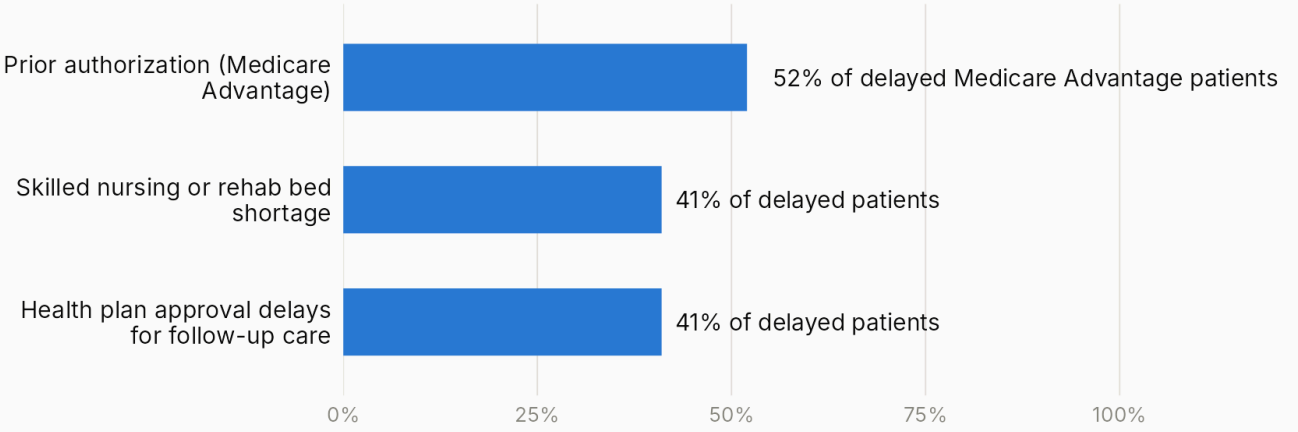
CHART 3 · PUBLISHED FIGURES

Why Florida patients stayed after discharge clearance

What drives discharge delay, and how many beds does it tie up?

Payer approvals and post-acute beds are the main listed causes of delay

Florida, July 2024 to June 2025: 243,000 patients stayed after clearance, 781,000 unnecessary days. That is 3.2 days each, about 2,140 beds occupied all year, and roughly $3,200 per delay day (my arithmetic from the survey totals).



Florida Hospital Association survey via The Capitolist (Jul 2026). A hospital association survey covering one state, not an independent check. Bases differ by bar, so the bars are not shares of one total.

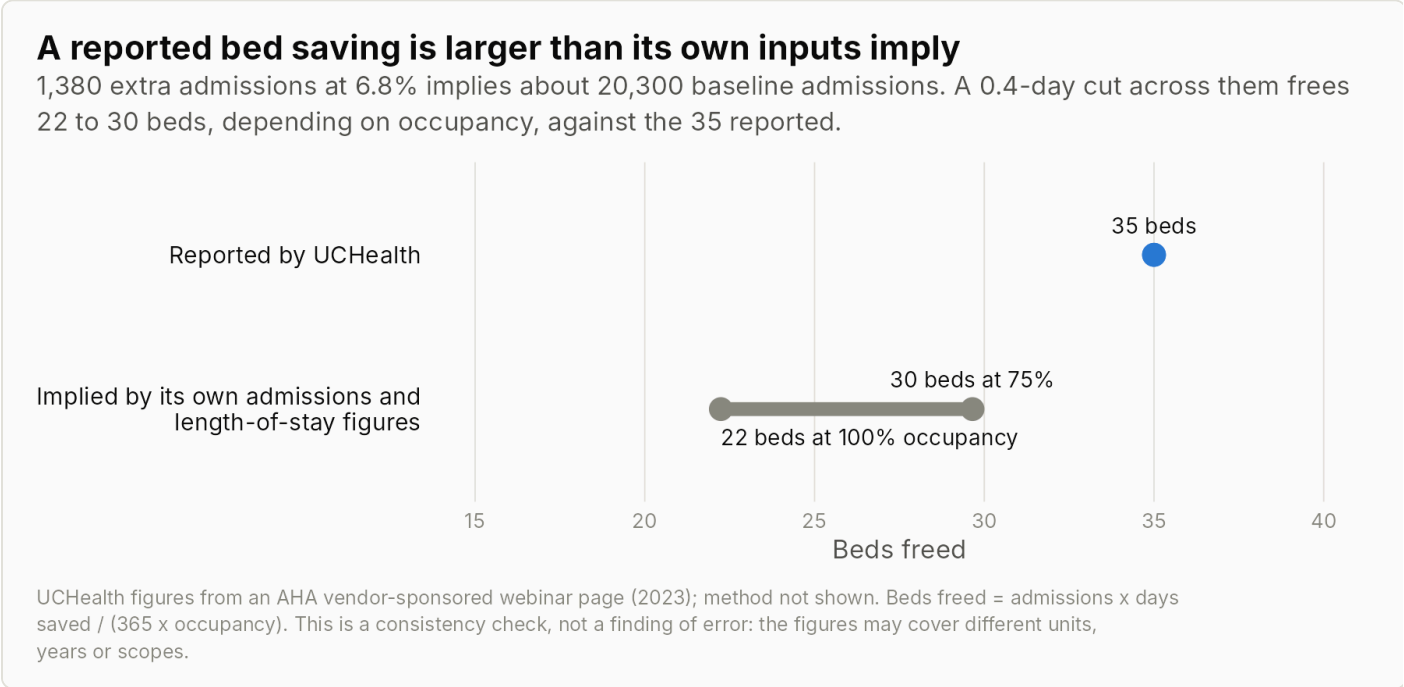
Dividing the survey totals gives 3.2 days per delayed patient and about $3,200 per delay day. 781,000 days is the same as about 2,140 beds occupied all year, which turns an abstract delay count into capacity.

> **Read with care.** The survey comes from a hospital association and covers one state. The bars use different bases, so they are not shares of one total, and the causes can overlap.

CHART 4 · PUBLISHED FIGURES

Does a reported bed saving match its own inputs?

Can a vendor-reported result be reproduced from the numbers published beside it?



Beds freed = admissions × days saved ÷ (365 × occupancy). The 1,380 extra admissions at 6.8% imply about 20,300 baseline admissions. A 0.4-day cut across them frees 22 to 30 beds at 100% to 75% occupancy, against 35 reported.

> **Read with care.** This is a consistency check, not a finding of error. The figures may cover different units, years or scopes, and the webinar page does not show its method.

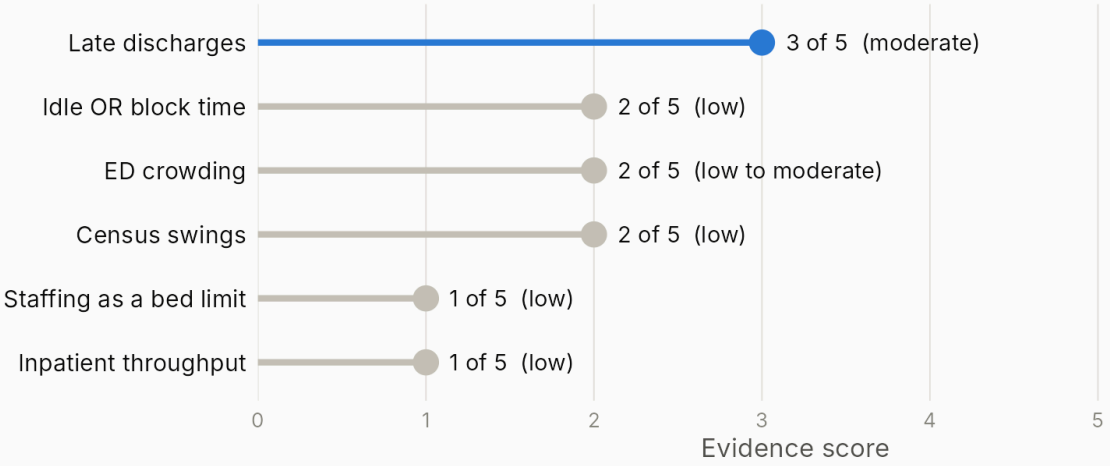
CHART 5 · PUBLISHED FIGURES

How strong is the evidence for each capacity use?

Which AI uses for capacity have the best evidence?

Discharge prediction has the strongest evidence, and it is only moderate

My 1 to 5 rating: 1 is an expectation or vendor claim, 5 is controlled, independent evidence. Ratings reflect study design and who reported the result.



Sources and reasoning for each row are in the written case study. Scores are my own judgement and are not from any source.

Discharge prediction scores highest at 3 of 5. Every other use scores 2 or below, because the results are self-reported, from a vendor, or from heterogeneous studies.

> **Read with care.** The score is my own rating of study design and who reported the result.

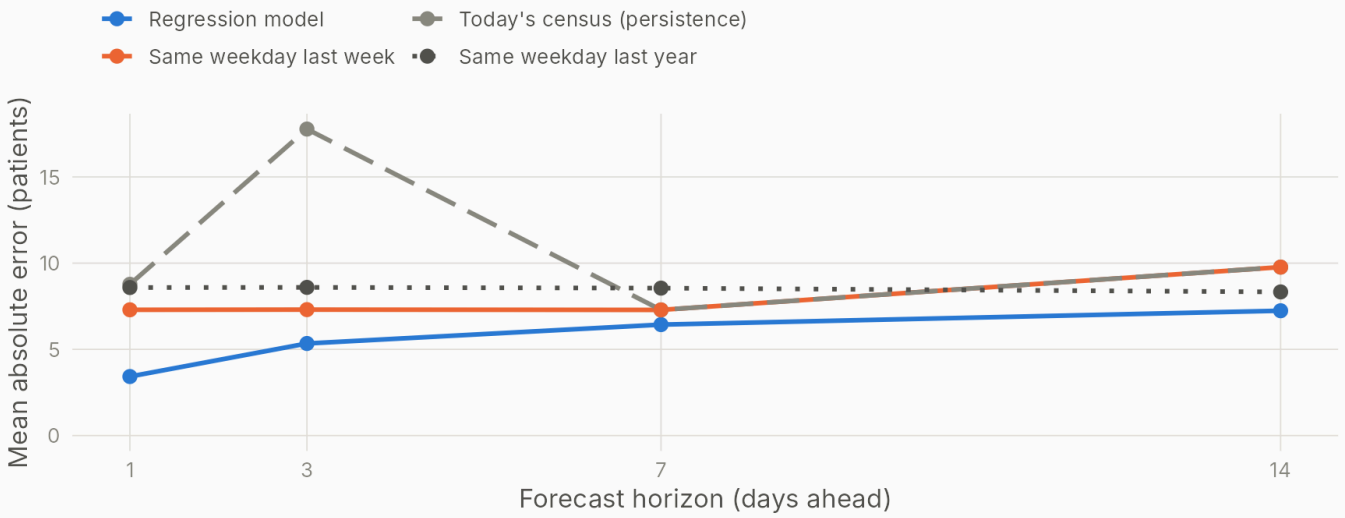
CHART 6 · SIMULATED DATA

Forecast error against simple baselines

How should a census forecast be evaluated?

Synthetic data: a forecast only counts if it beats the simple baselines

Mean absolute error in patients, rolling-origin evaluation over 2025 on simulated daily census for a 400-bed hospital. Skill versus the best baseline: 1-day 53%; 3-day 27%; 7-day 12%; 14-day 13%.



SIMULATED DATA. Shows how to evaluate a census forecast (same rows for every method, only information known at the forecast origin, baselines first). It says nothing about any real hospital. Skill = 1 - model MAE / best baseline MAE. At 7 and 14 days, persistence and same weekday last week are the same forecast, so their lines overlap.

Simulated daily census for a 400-bed hospital, evaluated with a rolling origin over 2025: each forecast uses only information available that day, the model is refit monthly, and every method is scored on the same rows. The model's mean absolute error is 3.4 patients one day ahead against 7.3 for the best baseline (53% skill), and 7.2 against 8.3 at 14 days (13% skill).

> **Read with care.** Simulated data. It shows the method, not how well any real forecast performs.

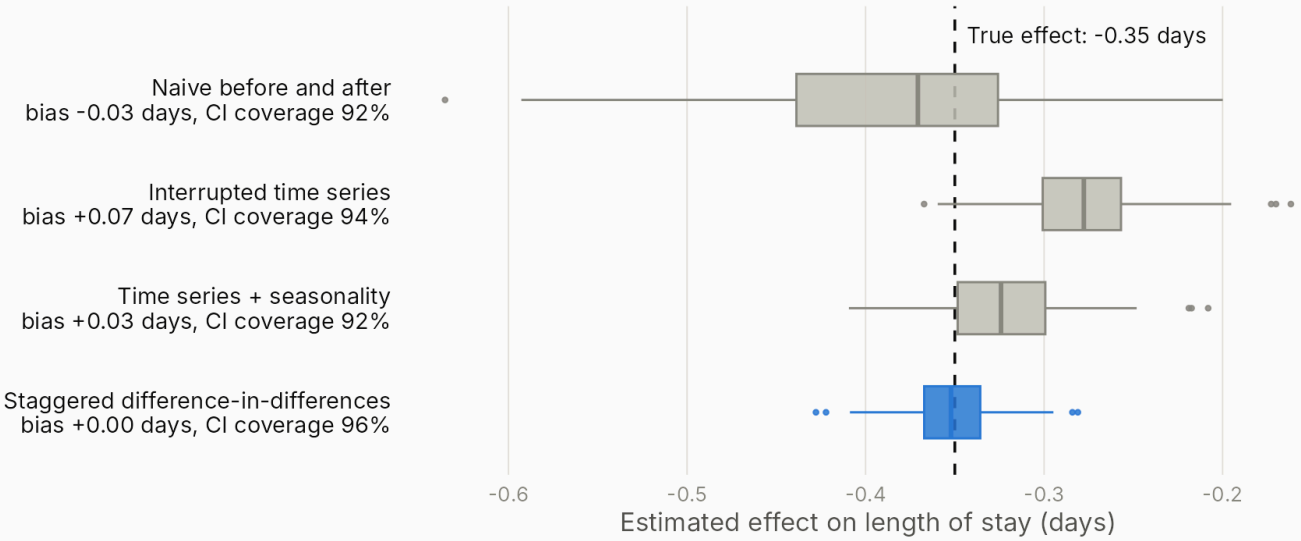
CHART 7 · SIMULATED DATA

Estimating an effect from a staggered rollout

Which estimator recovers an effect when units adopt a tool at different times?

Synthetic data: staggered difference-in-differences recovers the true effect

Estimated change in length of stay (days) across 200 simulated rollouts of 16 units over 104 weeks. A secular decline, winter seasonality and a hospital-wide shock confound the before-and-after comparisons.



SIMULATED DATA with a known effect of -0.35 days. Standard errors clustered by unit. Results depend on the simulation design: a different confounding pattern would change which simple methods fail. The naive estimate lands near the truth on average only because the downward trend and the upward shock roughly offset; its error is more than 3x larger. It shows why the measurement plan uses controls, not that any real tool works.

200 simulated rollouts across 16 units and 104 weeks, with a known effect of −0.35 days and three confounders: a secular decline, winter seasonality and a hospital-wide shock. Staggered difference-in-differences has the smallest error (RMSE 0.025 days against 0.084 for naive before and after) and 96% confidence-interval coverage. The naive estimate is close on average only because the trend and the shock roughly offset.

> **Read with care.** Simulated data. A different confounding pattern would change which simple methods fail.

Derived numbers

METRIC	VALUE	HOW IT IS COMPUTED
Implied bed change if census held flat	−14.7%	64% ÷ 75% − 1, against 16% reported
Demand growth implied by 75% to 85%	1.6% a year	$(85 \div 75)^{1/8} - 1$
Peak versus off-peak, 4+ hour waits	+10 points	35% − 25% (lower bound)
Days per delayed patient, Florida	3.2	781,000 ÷ 243,000
Beds occupied all year by delay	about 2,140	781,000 ÷ 365
Cost per delay day	about $3,200	$2.5 billion ÷ 781,000
UCHealth baseline admissions	about 20,300	1,380 ÷ 6.8%
Beds freed by a 0.4-day cut	22 to 30	20,294 × 0.4 ÷ (365 × occupancy), 100% to 75%

READ WITH CARE

Caveats

- The 1 to 5 evidence score is my own rating of study design and who reported the result.
- The UCLA occupancy study and the Health Affairs boarding analysis were read through secondary reports; the original papers were not opened.
- The Florida discharge-delay figures come from a hospital association survey covering one state, and the listed causes can overlap.
- The UCHealth check uses a vendor webinar page. Its figures may cover different units, years or scopes.
- Charts 6 and 7 and the simulation parameters are illustrative; they show how to evaluate a claim, not how well any real tool performs.